
PROJECT EXPO

TEAM MEMBERS:

DHARSHANA S

ELAKIYAA K

MARIYAM SHAMNA M

PRASHEETHA S

Smart IoT-Based Shopping Trolley Using RFID

PROBLEM STATEMENT:

In supermarkets, customers spend significant time waiting at billing counters, as manual barcode scanning increases checkout time and leads to long queues, especially on weekends and during festive seasons. Customers cannot monitor their bill while

shopping, making budgeting difficult, and often waste time searching for products due to the lack of in-store navigation. Existing trolleys lack automatic billing, item tracking, and theft prevention, with no way to verify that items in the trolley match the billed items before checkout, leaving them vulnerable to shoplifting. Store staff also lack real-time visibility into stock levels, causing stockouts, delayed restocking, and error-prone manual inventory audits. Additionally, dependency on billing counters raises operational costs due to extra staffing during peak hours, while the absence of digital payment options at the trolley adds to wait times. Elderly and differently-abled customers face further difficulty due to the lack of assistive features in manual scanning and heavy trolleys. Overall, the absence of a smart, integrated trolley system leads to inefficient checkout, poor inventory management, security risks, and a subpar shopping experience.

ABSTRACT:

The Smart IoT-Based Shopping Trolley is designed to automate the shopping experience using RFID technology and IoT. Every product carries an RFID tag. When a customer places an item in the trolley, the RFID reader detects the tag and automatically updates the product name, quantity, and price on an LCD display and cloud database. If an item is removed, the system automatically deducts it from the bill. The total bill is continuously displayed, allowing customers to track their expenses. At checkout, the final bill is sent to the billing server, eliminating the need for manual scanning. This system reduces billing time, minimizes human errors and improves the shopping experience.

OBJECTIVES:

1. To eliminate the need for cashier-assisted billing by enabling a self-checkout process directly through the trolley.
2. To reduce human error in billing caused by manual scanning, mismatched prices, or missed items.
3. To provide personalized product recommendations and offers to customers based on items added to the trolley.
4. To enable seamless digital/contactless payment options directly from the trolley, avoiding counter dependency.
5. To assist store management with real-time inventory tracking by linking trolley data with stock levels.
6. To reduce paper waste by offering digital bills/receipts instead of printed ones.

7. To improve accessibility for elderly and differently-abled customers through an easy-to-use trolley interface.
8. To enhance store security by preventing unbilled items from leaving the premises through automated verification.
9. To collect and analyze customer shopping pattern data to help supermarkets optimize product placement and staffing.
10. To build a low-cost, energy-efficient hardware model that can be easily adopted by small and medium-sized retail stores.

SCOPE OF THE PROJECT:

This project focuses on developing a smart trolley system integrated with RFID-based automatic billing, real-time bill tracking and in-store navigation assistance. The system is designed for deployment in supermarkets and retail stores to reduce checkout time, minimize manual errors, and enhance the overall shopping experience. It covers item detection

and billing as products are placed in the trolley, a display unit for real-time cost updates, and a basic navigation aid to help customers locate product sections. The project can be extended in the future to include features like digital payment integration, personalized recommendations, and inventory management, making it scalable for both small retail stores and large supermarket chains. However, the current scope is limited to prototype-level implementation and may not cover full-scale deployment across multi-floor or extremely large retail spaces.

EXPECTED OUTCOMES:

Faster checkout with reduced billing queues.

Accurate, error-free billing.

Real-time bill tracking for better budgeting.

Theft prevention through automated item verification.

Easy product navigation within the store.

Lower staffing costs for supermarkets.

Improved overall shopping experience.

Working prototype proving RFID-based trolley feasibility.

LITERATURE SURVEY:

S.No	Author(s) / Year	Title	Technology Used	Key Findings	Limitation / Gap
1	ANITS Project Report, 2019-20	RFID Based Smart Trolley for Automatic Billing System	RFID, Arduino, LCD	Customers scan products, view finalized bill in trolley, pay via customer card; data sent to central PC	No handling for products accidentally dropped into trolley; needs server maintenance Anits
2	Kulkarni et al., 2023	Smart Shopping Cart with Automated Billing Using Arduino	RFID tags + reader, Arduino	Automated billing within cart using RFID tags scanned as items added, reducing checkout queues Jetir	No theft-prevention or navigation feature
3	Shahroz et al., IEEE Access, Vol. 8	IoT Based Smart Shopping Trolley Using RFID	RFID sensors, Arduino, Bluetooth, mobile app	Efficiently managed shopping and billing, reducing waiting time and enhancing customer experience IRJMETS	No in-store navigation included
4	Jaishree et al., 2021	Smart Shopping Trolley Using IoT	Barcode scanner, LCD, Raspberry Pi	Reduced waiting time in queues during rush hours using barcode-based scanning IRJMETS	Barcode slower than RFID for bulk items
5	Karpagam V. et al., 2017	Smart Trolley with Smart Billing	RFID reader, LCD, ZigBee	Total cost of scanned products stored in microcontroller memory and	ZigBee has distance/range barrier IIETA

S.No	Author(s) / Year	Title	Technology Used	Key Findings	Limitation / Gap
6	Karthi et al., 2023	IoT Based Smart Trolley System for Automated Billing	RFID + obstacle sensors	displayed to customer IIETA Trolley followed customers automatically while avoiding obstacles for secure shopping Jetir	High hardware complexity/cost

EXISTING SOLUTIONS ANALYSIS:

S.No	Existing Solution	Technology Used	Advantages	Disadvantages
1	Traditional Barcode Billing (Manual Checkout)	Barcode scanner at counter	Low cost, widely used, simple to maintain	Long queues, time-consuming, requires staff, no real-time bill tracking for customer
2	Self-Checkout Kiosks	Barcode/QR scanning by customer	Reduces cashier dependency, faster than manual counters	Still requires manual scanning of each item; prone to scanning errors/theft; limited kiosks cause queues
3	RFID-Based Smart Trolleys (existing research)	RFID tags + RFID reader on trolley	Automatic item detection, faster billing, real-time bill display	No item-location/navigation feature; no verification for accidentally dropped items; needs infrastructure investment
4	Amazon Go / Just Walk Out Technology	Computer vision, sensor fusion, deep learning	No checkout needed at all; fully automated; highly accurate	Very high implementation cost; requires extensive camera/sensor network; not feasible for small/medium retailers
5	Mobile App-Based Scan & Go	Customer scans items via smartphone app	No extra hardware on trolley; flexible; customer-owned device used	Relies on customer honesty (self-scanning can be skipped); no automatic theft prevention; needs strong app adoption
6	ZigBee-Based Smart Carts	ZigBee + RFID	Wireless communication between cart and billing system	Limited range/distance; signal interference in crowded stores
7	IoT-Based Trolleys with Bluetooth/App Integration	RFID + Bluetooth + mobile app	Real-time billing sync with customer's phone; better UX	No standard theft-detection mechanism; dependent on stable Bluetooth connectivity

GAP ANALYSIS:

Existing systems	Proposed system
RFID billing only Manual scan or basic auto-bill	Automatic RFID billing No manual scanning needed
No live bill visibility Bill known only at checkout	Real-time bill display Updates as items are added
No theft verification Risk of unbilled items	Item cross-check at exit Trolley vs billed items
No in-store navigation Customers search manually	Section-wise navigation RFID/BLE zone mapping
High cost or limited range Amazon Go / ZigBee limits	Low-cost, scalable design Reuses existing RFID tags
Core gap No system unifies billing, security and navigation	

REFERENCES:

- M. Shahroz, M. F. Mushtaq, M. Ahmad, S. Ullah, A. Mehmood, and G. S. Choi, "IoT-Based Smart Shopping Trolley Using Radio Frequency Identification," *IEEE Access*, vol. 8, pp. 68426–68438, 2020.

- "Smart Shopping Trolley Using Arduino UNO," *IEEE Conference Publication*, 2024.
- R. Kulkarni, A. Kumavat, A. Jha, A. P. Singh, and A. Chincholkar, "Smart Shopping Cart with Automated Billing Using Arduino," 2023.
- S. Karthi, K. Sathiya Sri, A. Venimalathy, and T. Sindhu, "IOT Based Smart Trolley System for Automated Billing Using RFID," 2023.
- M. Jaishree, K. R. Lakshmi Prabha, S. Jeyaprabha, and K. Mohan, "Smart Shopping Trolley Using IOT," *2021 7th International Conference on Advanced Computing and Communication Systems*.
- P. Chandrasekar and T. Sangeetha, "Smart Shopping Cart with Automatic Billing System through RFID and ZigBee," *International Conference on Information Communication and Embedded Systems*.
- V. Karpagam, S. Balapriya, G. Kalairubini, and A. Kalaivani, "Smart Trolley with Smart Billing," *International Journal of*

Computer Systems, vol. 4, no. 3, pp. 55–58, 2017.

- A. Yewatkar, F. Inamdar, R. Singh, and A. A. Bandale, "Smart Cart with Automatic Billing, Product Information, Product Recommendation Using RFID & ZigBee," 2018.
- Kavitha T., Sasi Varun Kumar Reddy B., Surendra Goud E., and Ramya Sree G., "Smart Shopping Trolley with Auto Billing System Using RFID," Dept. of CSE, Srinivasa Ramanujan Institute of Technology, 2025.
- Priyanka Bhosale, "The RFID Based Smart Shopping Cart," *International Journal of Engineering Research and General Science*, vol. 3, no. 2, pp. 275–280, 2015.